

FMAA Cheat Sheet

Sections A and B, from the PRC e-books. Built for open-notes lookup: find the topic in the table of contents, go to the page. Pages 2–7 are the fast lane: the answer to every practice-quiz question and every formula.

Contents

Quiz answer finder (all 49 PRC quiz items)	2–3	A.3 Financial statements, cash flows	10
Formula sheet	4	A.4 Internal controls	11
Decision tables: which bucket?	5–6	A.5 Working capital, cash, A/R, inventory, A/P	12
Journal entries	7	B.1 Receivables, inventory, PP&E, liabilities	13
A.1 Terminology, business forms, principles	8	B.1 Revenue recognition, equity	14
A.2 Accounting cycle	9	B.2 Common-size analysis · B.3 Ratios	15

Reading this sheet

Green box	A rule of thumb or a worked calculation you can copy.
Orange box	A trap: the answer the exam expects that people get wrong.
QUC 1234	A real PRC quiz question number. If you recognize the question, the answer is right there.
Bold	The exact fact the question turns on.

The ten rules that answer the most questions

1	Order of the cycle: post → adjust → close → reverse . Reversing is optional and is the exact opposite of the adjusting entry.
2	Interest paid and interest/dividends received are operating . Dividends paid and treasury stock are financing . Buying or selling long-lived assets is investing .
3	Indirect method: add non-cash expenses and losses, subtract gains, subtract increases in operating assets, add increases in operating liabilities.
4	Current = within one year or the operating cycle, whichever is longer . A short-term debt with intent and ability to refinance is long-term.
5	Rising prices: FIFO gives lowest COGS, highest income, highest ending inventory. LIFO the opposite.
6	Revenue is recognized when control transfers , not when cash arrives. Cash before performance = liability (unearned revenue).
7	Cash dividend is a liability at declaration . Small stock dividend (under 20–25%) reduces retained earnings at market value ; large at par . Stock dividends and splits never change total equity.
8	Change in estimate (life, salvage, collectability, depreciation method) = prospective. Change in principle or an error = adjust opening retained earnings, net of tax.
9	Segregate authorization, recordkeeping, custody, reconciliation . Internal control gives reasonable , never absolute, assurance. Board is ultimately responsible; management maintains.
10	Vertical common-size base: total assets (balance sheet) or net sales (income statement). Horizontal: % of the base year; growth = index – 100.

Quiz answer finder

Every PRC practice-quiz question in Sections A and B, grouped by topic. Recognize the question, read the answer, check the reason.

Principles and the cycle

QUC	Question is about	Answer	Reason
12328	Feature unique to managerial accounting	Reports prepared based on management needs	Financial accounting must follow standards, report periodically, and looks at the past.
11942	Entity owned by 2+ firms for a special purpose	Joint venture	Owners share costs, returns and risks.
12322	Conservatism under uncertainty	Least likely to overstate assets (or income)	Also: least likely to understate liabilities or expenses.
317	Catalogues printed in Dec for a Jan promotion	Deferred cost, because of matching	They produce revenue in January, so the expense belongs in January.
314	When manufacturing labor hits the income statement	When the product is sold	Labor is capitalized into inventory first (matching).
2638	Order of cycle steps	Posting, adjusting, closing, reversing	Reversing happens at the start of the next period.
2235	Dr Prepaid Property Taxes / Cr Property Tax Expense	Recognizes a prepaid expense	Moves an unexpired cost out of expense into an asset.
3294	Accrue revenue from credit sales: effect	Assets ↑, liabilities no effect, equity ↑	Dr Receivable / Cr Revenue.
319	Reversing entries	Exact opposite of prior-period adjustments	Optional; made at the beginning of the next period.
12317	Which is a real (permanent) account	Allowance for doubtful accounts	Balance-sheet contra-asset. Losses, gains, expenses are temporary.

Financial statements and cash flows

QUC	Question is about	Answer	Reason
1751	What the cash flow statement does NOT assess	Management's efficient, profitable use of resources	That is the income statement's job.
6104	Buy asset \$5,000, borrow \$10,000	Investing outflow 5,000; financing inflow 10,000	
6106	Dividends 150k, interest 250k, equipment 350k: operating total	\$250,000	Only interest paid is operating.
1752	Purpose of the statement of financial position	Evaluate economic resources and obligations	Performance is the income statement; cash flows are the CFS.
1881	Income 60,000 incl. depreciation 5,000, bond-discount amortization 6,000, interest 4,000: cash flow	\$71,000	Add back the two non-cash items. Interest was paid in cash.
325	Net income of a proprietorship from balance-sheet changes	\$12,000	End equity 99,000 = Beg 90,000 + 72,000 invested – 75,000 withdrawn + NI.
1960	Which is a financing activity	Purchase of treasury stock	Interest on a mortgage is operating; equipment and trademarks are investing.
1475	Credit-card interest incurred and paid	Equity decreases; operating outflow	Interest expense reduces income.
2507	\$500,000 employee inventory theft, no restitution	Loss, separate line on the income statement	Material fraud is not buried in COGS.
2793	Which item is a change in estimate	Change in collectability of receivables	Completed-contract → percentage-of-completion is a change in principle.

Internal control and working capital

QUC	Question is about	Answer	Reason
6450	Ultimate objective of managerial control	Promote organizational effectiveness	Control increases the likelihood that objectives are achieved.
7645	Cost ≤ benefit; controls should not hinder efficiency	Reasonable assurance	Absolute assurance would cost too much.
7646	Which cash procedure violates control	Cashier prepares a weekly deposit slip for all receipts	Custody plus recording in one person, and not timely.
7763	Best sign management is committed to controls	Funds allocated annually to improve information systems	Aggressive reporting or earnings pressure are bad signs.
128	Risk that controls fail to prevent/detect/correct errors timely	Control risk	Inherent = absent controls; detection = auditors miss it.
12090	Which combination decreases net working capital	Cash ↓ \$1M and fixed assets ↑ \$1M	Current assets fall, current liabilities unchanged.
6189	Not a motive for holding cash	Motive to make a profit	Transaction, precautionary, speculative (future needs) are the motives.

QUC	Question is about	Answer	Reason
6214	Not a carrying cost of inventory	Shipping costs	Storage, insurance, opportunity cost are carrying costs.
6882	Best financing for a new factory machine	Retained earnings or a long-term loan	Maturity matching: long-lived asset, long-term money.
589	Factoring means	Selling accounts receivable to another company	

Recognition, valuation, equity

QUC	Question is about	Answer	Reason
7398	A/R on May 1 with 60/25/15 collection pattern; Mar 280k, Apr 260k	\$146,000	$15\% \times 280,000 + 40\% \times 260,000$.
12113	FIFO ending inventory, 11,500 units, purchases at \$7.50	\$86,250	All ending units come from the latest purchases: $11,500 \times 7.50$.
7000	SYD, cost 250k, salvage 50k, 5 years: year-3 expense and accumulated	\$40,000 and \$160,000	$200,000 \times 3/15$; $200,000 \times 12/15$.
1478	Selling price when cost 700k, accum. dep. 495k, gain 75k	\$280,000	Book value $205,000 + \text{gain } 75,000$.
546	Short-term notes excludable from current liabilities	\$340,000 (both notes)	One refinanced, one under a non-cancelable agreement, both before issuance.
12115	5% stock dividend, 100,000 shares, \$10 par, \$18 market: RE effect	Decrease \$90,000	Small dividend: $5,000 \text{ shares} \times \18 market .
11810	Revenue in year 1 when 50,000 of 100,000 gallons delivered at \$3	\$150,000	Delivery transfers control; cash timing is irrelevant.
525	Annual lawn-service fee paid in advance	Over the year as services are performed	Each visit is a distinct performance obligation.
11813	Cash dividend distributed + 5% stock dividend declared: equity change	\$0	Cash dividend already left equity at declaration; stock dividend moves within equity.

Analysis and ratios

QUC	Question is about	Answer	Reason
2419	Base of a common-size balance sheet	Total assets	Income statement uses net sales.
6910	Example of vertical common-size analysis	Advertising is 2% of sales this year	Item ÷ same-year base.
12057	Common-size % for net income 120,000, sales 600,000	20%	
12059	Largest % change: sales 6.0 → 6.6M, NI 500 → 540k, assets 10 → 10.5M, inventory 600 → 500k	Inventory	$500/600 = 83\%$, a 17% change; the others are 10%, 8%, 5%.
12182	Growth rate of salaries 568,920 → 622,310	9.38%	$622,310 \div 568,920 = 1.0938$.
6026	A/R turnover: credit sales 3.6M; A/R 320k → 400k	10.0	$3,600,000 \div \text{average } 360,000$.
6068	Return on equity: NI 96,000; equity 467,000 → 534,000	19.2%	$96,000 \div 500,500$. Include the reserve for bond retirement; exclude bonds.
140	Ratio whose unexpected decrease suggests fictitious inventory	Total assets turnover	Fake inventory inflates assets. Current ratio would rise.
11968	Inventory when CL 2,000,000, NWC 300,000, quick ratio 0.8	\$700,000	$CA = 2,300,000$; $(2,300,000 - \text{Inv}) \div 2,000,000 = 0.8$.
11969	Times interest earned: EBIT 20M, interest 5M	4	$\text{EBIT} \div \text{interest}$.

Formula sheet

Liquidity

Current ratio	$\frac{\text{Current assets}}{\text{Current liabilities}}$
Quick (acid-test)	$\frac{\text{Cash} + \text{Marketable securities} + \text{Receivables}}{\text{Current liabilities}}$
Cash ratio	$\frac{\text{Cash} + \text{Marketable securities}}{\text{Current liabilities}}$
Cash flow ratio	$\frac{\text{Cash flow from operations}}{\text{Average current liabilities}}$
Net working capital	$\text{Current assets} - \text{Current liabilities}$
NWC ratio	$\frac{\text{Net working capital}}{\text{Total assets}}$

Leverage

Debt to assets	$\frac{\text{Total debt (CL + long-term debt)}}{\text{Total assets}}$
Debt to equity	$\frac{\text{Total debt}}{\text{Common equity}}$
LT debt to equity	$\frac{\text{Total debt} - \text{Current liabilities}}{\text{Common equity}}$
Times interest earned	$\frac{\text{EBIT}}{\text{Interest expense}}$

Lenders prefer high TIE with high debt over low TIE with low debt. ROE above ROA means leverage is paying off.

Activity (360-day year)

Inventory turnover	$\frac{\text{Cost of goods sold}}{\text{Average inventory}}$
Days in inventory	$360 \div \text{Inventory turnover}$
Receivables turnover	$\frac{\text{Net credit sales}}{\text{Average receivables}}$
Days sales outstanding	$360 \div \text{Receivables turnover}$
Payables turnover	$\frac{\text{Net credit purchases}}{\text{Average payables}}$
Days payables	$360 \div \text{Payables turnover}$
Operating cycle	$\text{Days inventory} + \text{Days receivables}$
Cash conversion cycle	$\text{Days inventory} + \text{Days receivables} - \text{Days payables}$
Asset turnover	$\frac{\text{Sales}}{\text{Average total (or fixed) assets}}$

Profitability

Gross margin	$\frac{\text{Net sales} - \text{COGS}}{\text{Net sales}}$
Operating margin	$\frac{\text{Operating income}}{\text{Net sales}}$
Net margin	$\frac{\text{Net income}}{\text{Net sales}}$
Return on assets	$\frac{\text{Net income}}{\text{Average total assets}}$
Return on equity	$\frac{\text{Net income} - \text{Preferred dividends}}{\text{Average common equity}}$

Depreciation (cost C, salvage S, life n)

Straight-line	$\frac{C - S}{n} \text{ per year}$
Activity	$\frac{C - S}{\text{Total capacity}} \times \text{units or hours used}$
Sum-of-years'-digits	$(C - S) \times \frac{\text{remaining life}}{n(n+1) \div 2}$
Double-declining	$\text{Book value} \times \frac{2}{n} \text{ (ignore S until floor)}$

Example, C 1,000,000, S 100,000, 5 yr: SL 180,000/yr · SYD 300k, 240k, 180k, 120k, 60k · DDB 400k, 240k, 144k, 86.4k, 29.6k · Activity 45/hr.

Other calculations

Periodic COGS	$\text{Beginning inv.} + \text{Purchases} - \text{Purchase returns} - \text{Ending inv.}$
Bad debt (aging)	$\text{Required allowance} - \text{Existing balance}$
Bad debt (% sales)	$\% \times \text{Credit sales (ignore existing balance)}$
Gain/loss on disposal	$\text{Proceeds} - (\text{Cost} - \text{Accumulated depreciation})$
Equity roll-forward	$\text{Beg} + \text{Investments} - \text{Withdrawals} + \text{Net income} = \text{End}$
Dividends (from RE)	$\text{Net income} - \text{Change in retained earnings}$
Growth rate	$(\text{Current} \div \text{Prior}) - 1$
Expected value	$\Sigma (\text{probability} \times \text{outcome})$
Allocate price	$\text{Contract price} \times (\text{SSP of item} \div \Sigma \text{SSP})$
Book value / share	$(\text{Assets} - \text{Liabilities} - \text{Preferred}) \div \text{Common shares}$
Value of a right	$(\text{Price rights-on} - \text{Subscription}) \div (\text{Rights per share} + 1)$

Decision tables: which bucket?

Cash flow statement

Operating	Investing	Financing
Cash from customers	Buy / sell PP&E	Issue stock
Paid to suppliers, employees	Buy / sell intangibles	Buy treasury stock
Taxes paid	Buy / sell other firms' stock or bonds	Borrow
Interest paid	Loans made to others and their collection (principal)	Repay principal
Interest received		Dividends paid
Dividends received		

Indirect method: net income → cash from operations

Add	Subtract	Ignore
Depreciation, amortization, impairment	Gains on disposals or debt	Interest paid in cash
Bond-discount amortization	Increase in receivables, inventory, prepaids	Asset purchases (investing)
Losses on disposals or debt	Decrease in payables, accruals	Loan repayment (financing)
Decrease in operating assets	Undistributed equity-method income	
Increase in operating liabilities		

Normal balances

Debit balance	Credit balance
Assets	Liabilities
Expenses, losses	Equity, retained earnings
Owner drawings	Revenues, gains
Treasury stock (contra-equity)	Accumulated depreciation (contra-asset)
Discount on notes payable (contra-liability)	Allowance for doubtful debts (contra-asset)
Sales returns, sales discounts (contra-revenue)	Premium on notes payable (adjunct)
	Sales discount forfeited (other revenue)

Permanent vs temporary

Permanent (real): carried forward	Temporary (nominal): closed to zero
All assets, incl. contra-assets	Revenues, gains
All liabilities	Expenses, losses
Capital stock, APIC	Dividends, drawings
Retained earnings	Income summary
Accumulated OCI	

Balance sheet classification

Current	Non-current
Cash, marketable securities, receivables, inventory, prepaids	Long-term investments, PP&E, intangibles, goodwill
Payables, accruals, unearned revenue due within a year	Cash restricted for a long-term purpose
Current portion of long-term debt	Bonds, long-term notes, leases, pensions, deferred taxes
	Short-term debt with intent and ability to refinance

Current = within one year or the operating cycle, whichever is longer.

Events after the balance-sheet date

Adjust the statements	Disclose only
Evidence about a condition that existed at the balance-sheet date (customer already failing; lawsuit already pending)	Condition arose after : stock issue or split, business combination, fire or flood, litigation from a later event, receivable losses from later conditions

Income statement: where does it go?

Item	Treatment
Unusual or infrequent gains/losses (write-downs, restructuring, FX, disposals, strikes, disasters)	Other revenues/expenses, before tax
Discontinued operations (whole segment or product line)	Separate section, net of tax
Employee theft of inventory	Loss, separate line
Change in estimate	Prospective; no restatement
Change in principle or error correction	Opening retained earnings, net of tax

Revenue: over time or point in time?

Over time (any one)	Point in time (control indicators)
Customer consumes as you perform (services, support)	Right to payment
You build or improve an asset the customer controls	Legal title passed
No alternative use and enforceable right to payment for work done	Physical possession
	Risks and rewards transferred
	Customer acceptance

Inventory: whose is it? What cost?

Situation	Answer
Goods on consignment	Consignor's inventory (plus freight to consignee)
In transit, FOB shipping point	Buyer's (title passes at the carrier)
In transit, FOB destination	Seller's until the buyer's dock
Freight-in, insurance in transit, DM, DL, factory overhead	Product cost: capitalize
Freight-out, selling, G&A, general interest	Period cost: expense

Dividends and splits

Event	Retained earnings	Total equity
Cash dividend declared	↓ (liability created)	↓
Cash dividend paid	no change	no change
Small stock dividend (<20–25%)	↓ at market value	no change
Large stock dividend (>20–25%)	↓ at par value	no change
Stock split	memo entry only	no change
Treasury stock purchase	no change	↓

Journal entries

Everyday transactions

Owner invests cash	Dr Cash	
	Cr Capital	
Prepay rent	Dr Prepaid Rent	
	Cr Cash	
Buy equipment for cash	Dr Equipment	
	Cr Cash	
Cash received in advance	Dr Cash	
	Cr Unearned Revenue	
Buy supplies on credit	Dr Supplies	
	Cr Accounts Payable	
Land: part cash, part note	Dr Land	
	Cr Cash	
	Cr Notes Payable	
Pay down a note	Dr Notes Payable	
	Cr Cash	
Service for cash	Dr Cash	
	Cr Revenue	
Service on account	Dr Accounts Receivable	
	Cr Revenue	

Adjusting entries

Rent used up	Dr Rent Expense	
	Cr Prepaid Rent	
Supplies used	Dr Supplies Expense	
	Cr Supplies	
Advance now earned	Dr Unearned Revenue	
	Cr Revenue	
Revenue earned, unbilled	Dr Accounts Receivable	
	Cr Revenue	
Expense incurred, unpaid	Dr Wages Expense	
	Cr Wages Payable	
Depreciation	Dr Depreciation Expense	
	Cr Accumulated Depreciation	
Bad debt estimate	Dr Bad Debt Expense	
	Cr Allowance for Doubtful Debts	

Closing entries (BestPort)

Revenue	9,000	
Income Summary		9,000
Income Summary	1,200	
Rent Expense		1,000
Supplies Expense		200
Income Summary	7,800	
Retained Earnings		7,800

After closing: revenue, expense and income summary = 0. Retained earnings carries the 7,800.

Receivables

Write off an account	Dr Allowance for Doubtful Debts	
	Cr Accounts Receivable	
Recover a written-off acct	Dr Accounts Receivable	
	Cr Allowance	
	Dr Cash	
	Cr Accounts Receivable	
Gross method, paid in time	Dr Cash	1,960
	Dr Sales Discount	40
	Cr A/R	2,000
Net method, paid late	Dr A/R	60
	Cr Sales Discount Forfeited	60
Factor without recourse	Dr Cash	
	Dr Loss on Sale of Receivables	
	Cr Accounts Receivable	

Assets and revenue

Sell asset at a gain	Dr Cash	
	Dr Accumulated Depreciation	
	Cr Asset (cost)	
	Cr Gain on Disposal	
Sell asset at a loss	Dr Cash	
	Dr Accumulated Depreciation	
	Dr Loss on Disposal	
	Cr Asset (cost)	
Deliver goods paid earlier	Dr Unearned Revenue	
	Cr Sales Revenue	
	Dr Cost of Goods Sold	
	Cr Inventory	
Sale with financing >1 yr	Dr A/R	13,310
	Cr Unearned Interest	3,310
	Cr Sales Revenue	10,000

Equity

Issue 1,000 sh, \$50 par @\$75	Dr Cash	75,000
	Cr Common Stock	50,000
	Cr APIC	25,000
Declare cash dividend	Dr Retained Earnings	
	Cr Dividends Payable	
Pay it	Dr Dividends Payable	
	Cr Cash	
Small stock dividend	Dr Retained Earnings (market)	
	Cr Common Stock (par)	
	Cr APIC (excess)	
Large stock dividend	Dr Retained Earnings (par)	
	Cr Common Stock (par)	
Buy treasury stock	Dr Treasury Stock	
	Cr Cash	
Stock split	Memo entry only	

A.1 Terminology, business forms, principles

Three kinds of accounting

Kind	Who uses it	Rules	Time focus
Financial	External: creditors, shareholders, investors, lenders, government	GAAP or IFRS; change over time; differ by country	Past; reports each period
Managerial	Internal decision makers (plan, evaluate, control)	No mandatory standards ; similar across countries; stable	Past and future; reports when management needs them
Tax	Tax authorities	Local tax law	Outside the FMAA

ICMA definition: accounting = identifying, classifying, measuring, recording and communicating, in monetary terms, financial transactions and events of an entity.

Forms of business

Form	Separate legal entity?	Liability	Remember
Sole proprietorship	No	Unlimited	Ends with the owner; one owner keeps all profit and loss
General partnership	No	All partners unlimited	Easy to form, even without meaning to; any change = new partnership
Limited partnership	By statute (certificate filed)	General partners unlimited; limited partners limited	Needs at least one of each; only general partners manage ; a limited partner's name cannot appear in the firm name
LLP	Statutory filing	Limited, except your own malpractice	Law, audit and similar professions; all partners may manage
Corporation	Yes	Limited	Unlimited life; shareholders elect the board, board appoints management; double taxation ; public ones regulated by SEC and PCAOB
S corporation	Yes	Limited	No corporate-level tax; at most 75 shareholders, one class of stock
LLC	Limited liability plus preferential tax treatment, with conditions		
Joint venture	Two or more parties, one specific purpose, shared costs, returns and risks		
Nonprofit	Public-benefit purpose, usually tax exempt, surplus reinvested rather than distributed		

Accounting equation and element definitions

Assets (debit) = Liabilities (credit) + Equity (credit)

Asset	Probable future economic benefit obtained or controlled as a result of past transactions
Liability	Probable future sacrifice of economic benefits from a present obligation, from past transactions
Equity	Residual interest in assets after liabilities: contributed capital plus retained earnings
Revenue / expense	Inflows / outflows from the entity's ongoing major or central operations
Gain / loss	Equity changes from peripheral or incidental transactions (a laptop maker selling its own machine)
Journal / ledger	Journal = book of original entry, in date order. Ledger = book of final entry. General ledger = every account.

Principles

Accrual basis	Revenue when earned, expense when incurred, whatever the cash timing. GAAP and IFRS require it; cash basis is tolerated only for small firms. Cash received before delivery is a liability .
Conservatism	Under uncertainty, book expenses and liabilities early, revenues and assets only when assured. Expense the ad campaign now; accrue the probable lawsuit; wait for the contract revenue.
Consistency	Same policies period to period; change only for a good, permanent reason.
Matching	Expense in the same period as the revenue it produced: COGS when sold, depreciation over the life, warranty and bad-debt estimates in the year of sale.
Accrual vs deferral	Accrual: earned or incurred, cash not yet moved. Deferral: cash moved, belongs to a later period.

A.2 Recording transactions: the accounting cycle

The ten steps, in order

1	Identify and measure the transaction in money
2	Journalize: date, debit(s), credit(s), equal amounts, description
3	Post to ledger accounts (T-accounts)
4	Unadjusted trial balance: proves debits = credits; not a financial statement ; cannot catch a wrong-account error
5	Adjust for accruals and deferrals
6	Adjusted trial balance
7	Statements: income statement → changes in equity → balance sheet → cash flows
8	Close temporary accounts through Income Summary into Retained Earnings
9	Post-closing trial balance: permanent accounts only
10	Reverse (optional): first day of next period, exact opposite of an adjusting entry

Tested order: **posting** → **adjusting** → **closing** → **reversing**.

Double-entry rules

- Every entry has at least one debit and one credit, and the totals are equal.
- Assets, expenses, losses grow with debits. Liabilities, equity, revenues, gains grow with credits.
- **Contra** account: opposite balance to the account it reduces (Accumulated Depreciation is a credit inside assets).
- **Adjunct** account: same balance as the account it adds to (Premium on Notes Payable is a credit).

Four adjusting entries

Kind	Entry
Accrued revenue	Dr Receivable / Cr Revenue
Accrued expense	Dr Expense / Cr Payable
Prepaid expense used	Dr Expense / Cr Prepaid asset
Unearned revenue earned	Dr Unearned Revenue / Cr Revenue

Permanent vs temporary

Permanent: every asset, liability and equity account, including contra-assets.

Temporary: revenues, expenses, gains, losses, drawings or dividends, Income Summary.

BestPort Company: the numbers the exam recycles

Checkpoint	Total	Why
Unadjusted trial balance	87,300	Debits = credits after ten December entries
Adjusted trial balance	87,300	Adjustments move amounts between accounts; totals unchanged
Net income	7,800	Revenue 9,000 – rent 1,000 – supplies 200
Total assets = liabilities + equity	86,100	Cash 51,500 + A/R 2,000 + supplies 600 + prepaid 2,000 + equipment 10,000 + land 20,000
Post-closing trial balance	86,100	Only permanent accounts remain; retained earnings 7,800

A.3 Financial statements and cash flows

Balance sheet (point in time)

Helps assess	Operating capability; capital structure, risk, creditworthiness; enterprise value; future cash flows; liquidity (time to turn into cash) and financial flexibility (ability to change amounts and timing of cash flows)
Limitations	Historical cost, not fair value; estimates and judgments; omits things that cannot be measured (people); snapshot only
Asset order	Current (cash, securities, receivables, inventory, prepaids) → long-term investments → PP&E → intangibles → other assets
Equity parts	Capital stock at par · additional paid-in capital · retained earnings (free and appropriated) · accumulated OCI · less treasury stock. Disclose shares authorized, issued, outstanding .
Disclosure	Parenthetical notes, footnotes, cross-references and contra items, supporting schedules. Supplemental: contingencies, accounting policies, contracts and covenants, post-balance-sheet events.

Statement of changes in equity

Beginning ± prior-period adjustment + net income + shares issued – dividends declared = ending. Net income comes in from the income statement; ending equity goes to the balance sheet.

Income statement (period)

Sales – discounts – returns = Net sales – Cost of sales = Gross profit – Selling expenses – G&A = Operating income + Other revenues and gains – Other expenses and losses = Income before tax – Income tax = Income from continuing operations ± Discontinued operations (net of tax) = Net income	
Helps assess	Past performance; future cash flows and their risk; sustainable earnings; management's efficient and profitable use of resources
Limitations	Depends on methods and estimates (quality of earnings); past only; leaves out items that cannot be measured reliably (some unrealized gains go to OCI; brand value)
Single vs multiple step	Single: revenues minus expenses. Multiple: separates operating from non-operating and shows gross profit and operating income.

Statement of cash flows

Purpose	Cash receipts and payments in three groups plus the net change in cash. Shows liquidity, solvency (operating cash flow is the key), financial flexibility.
Limitations	Leaves out non-cash transactions; classification takes judgment; historical.
Built from	Comparative balance sheets, the current income statement, selected transaction data.
Method	Only the indirect method is tested: start with net income, adjust for non-cash items and working-capital changes. Direct method is rare because firms do not keep cash-receipt and cash-payment records by category.

Worked bridge: Net income 40,000 + depreciation 18,000 – gain on machine 4,000 + decrease in A/R 5,000 – increase in inventory 10,000 – increase in prepaids 3,000 – decrease in A/P 8,000 = **cash from operations 38,000**. Investing: +18,000 sale – 60,000 purchase = –42,000. Financing: +41,000 stock – 20,000 loan – 10,000 dividends = +11,000. Cash 20,000 → 27,000.

Hidden numbers: depreciation expense = change in accumulated depreciation + accumulated depreciation removed on a disposal (cost – book value). Dividends paid = net income – change in retained earnings.

Interest paid is operating even though the loan is financing. Dividends received are operating; dividends paid are financing. Buying treasury stock is financing.

A.4 Internal controls

Governance and control

Governance	The system by which an organization is directed and controlled. Goal: long-term shareholder value while balancing other stakeholders. Risk management and control are components of it.
Agency problem	Owners (principals) hire managers (agents) whose interests differ; the board monitors management.
Responsibility	Board: ultimate responsibility. Senior management: establishes and maintains the system.
COSO definition	A process, effected by board, management and personnel, giving reasonable assurance of: (1) effective and efficient operations, (2) reliable financial reporting, (3) compliance with laws.
Reasonable assurance	Never absolute. Collusion, management override and cost-benefit limits mean controls can fail. Cost must not exceed benefit.
Ultimate objective	Organizational effectiveness: achieving objectives and goals.

Strong vs weak controls

Strong system gives	Weak system risks
Better decisions from reliable data Protected assets Lower external audit cost (auditor relies on controls)	Employee theft Loss of critical information Inefficiency, poor decisions Higher audit fees

Segregation of duties: four jobs, four people

Authorization	Recordkeeping	Custody	Reconciliation
Approve credit, sign purchase orders, set pay rates	Journal entries, posting payments, payroll preparation	Hold cash, checks, securities, inventory	Bank reconciliation, physical counts vs records

- A/R adjuster ≠ payment poster. Cash handler ≠ write-off approver. Deposit preparer ≠ bank reconciler.
- Purchase-order issuer ≠ receiving recorder (stops fictitious-vendor fraud).
- IT: separate requesting, authorizing, developing, reviewing, implementing a change.
- Segregation does not stop **collusion**. When separation is impossible, use **compensating controls**: audit trails, application reconciliation, exception reports reviewed by a supervisor, transaction logs, supervisory review.

Safeguarding assets

Cash and payments	Physical access	Logical access
Registers, lockbox, intact daily deposits, treasury custody Pre-numbered forms Three-way match : purchase order + invoice + receiving report, then cancel the voucher Credit memos and write-offs approved independently Payroll from approved time cards; checks handed out by treasury; unclaimed checks held independently	Locks (metal, electronic, biometric) Deadman doors, alarms, cameras, guards Color-coded visitor badges, escorted visitors Single entry point Bonded employees	Unique passwords, biometrics Encryption, firewalls, anti-malware Backup and recovery Secure PC storage

Compare records with assets using an **independent person**; frequency depends on value and cost (cash daily, inventory less often).

Risk vocabulary

Responses	Accept · Treat (reduce, usually with controls) · Transfer · Avoid
Inherent → residual	Inherent (gross) risk before controls; managed risk is mitigated; residual (net) risk remains and is what profit rewards.
Inherent risk (reporting)	Susceptibility to misstatement with no controls
Control risk	Controls fail to prevent, detect or correct in time
Detection risk	Auditors or checkers fail to find the misstatement
Control-risk drivers	Control consciousness · design (segregation) · communication · consistent application · independent checks · monitoring

Structure and management style

- Centralized: standardized, easy to enforce. Decentralized: varied, hard to make uniform.
- Rigid hierarchy (army): least flexible, least risky, most control. Flat and free: flexible, risky.
- Commitment shows in: money for information systems, use of budgets and standards, attitude to auditors, open-door contact with staff. Warning signs: a small clique decides everything, aggressive reporting, pressure to hit earnings targets.

A.5 Managing daily finances

Working capital

Net working capital = Current assets – Current liabilities

Goal: shorten the operating cycle (buy → hold → sell → collect) and the cash cycle (operating cycle minus days you take to pay suppliers). More liquid assets earn less; shorter-term debt costs less.

Policy	Financing	Result
Aggressive	Short-term liabilities fund longer-term assets	Highest return, risk of insolvency
Conservative	Long-term money funds even current assets	Safest, lowest return
Maturity matching	Match asset life to financing term	Middle ground; a machine is financed with retained earnings or a long-term loan

Cash

Why hold it	Transaction, precautionary, speculative. Profit is not a motive; cash earns the least.
How much	Depends on revenue pattern and seasonality, operating expenses, contingencies, growth opportunities, market conditions.
Forecast says	Deficit → arrange short-term financing early. Surplus → invest part of it.

Receivables

Credit policy	Set by top management, run by the credit department: credit period, credit standards, limits, discount rate and period. Balance extra sales against default (credit) risk.
Loosen it	Sales up, receivables up, bad debts and collection costs up, opportunity cost up. A cash discount speeds collection.
Factoring	Selling receivables below face value. With recourse: you still cover defaults; a sale only if you give up control, the recourse obligation can be estimated, and the factor cannot force a repurchase; otherwise it is a loan. Without recourse: factor bears the risk, costs more, treated as a sale with a loss.

Inventory

Types	Raw materials (traceable) · work in process · finished goods · materials and supplies (not traceable)	
Why hold it	Supply and demand mismatch, production delays, quantity discounts, level operations, avoid stockouts	
Two questions	When to order? How much?	
Ordering costs	Carrying costs	Stockout costs
Preparing and placing orders; discounts lost by ordering too little	Storage, freight-in and insurance on shipment, facility rent and depreciation, insurance, property tax, security, spoilage, theft, obsolescence, opportunity cost of the money	Lost sales and goodwill. Watch lead time: order placed → goods ready to use

Shipping goods out to customers is **not** a carrying cost.

Just-in-time	Inventory arrives as needed; demand driven; no warehousing. Cuts carrying cost, theft, control effort, property tax; needs reliable suppliers (small safety stock); exposes waste and defective parts. Less carrying cost, more stockout cost.
MRP	Computer system that turns the master production schedule into purchase and production schedules; tracks stock, orders, lead times, usage; fewer rush orders.

Payables

Trade credit is borrowing; use it only when cheaper than other money (cost depends on the discount you give up). Ten-step cycle: requisition → **purchase order (legally binding)** → receive and check goods (receiving report) → vendor invoice → **three-way match** → approval → record liability → authorize payment → pay → reconcile and confirm. Benefits: trade credit, no late fees, tracking, fraud prevention (duplicate invoices), smoother cash flow, supplier goodwill, better reporting.

B.1 Receivables, inventory, PP&E, intangibles, liabilities

Receivables

Valuation	Short-term at net realizable value ; noncurrent at present value . Recorded net of trade discounts. Notes receivable are written promises. Nontrade receivables (employee advances, dividends receivable) shown separately.
2/10, n/30	2% off if paid in 10 days, else full amount in 30. Gross method: book full amount, record Sales Discount when taken. Net method: book at 98%; a late payer produces Sales Discount Forfeited (other revenue).
Allowance, % of sales	Expense = % × credit sales. Existing allowance balance is ignored. $1\% \times 10,000,000 = 100,000$.
Allowance, aging	Compute the required ending allowance, then expense = required – existing. Aging $420k \times 3\% + 18k \times 10\% + 14k \times 30\% + 50k \times 75\% = 56,100$; less 12,500 on hand = 43,600 .
Roll-forwards	A/R: Beg + credit sales – collections – write-offs = End. Allowance: Beg + expense + recoveries – write-offs = End.

Inventory

Periodic	Purchases account all year; count at year end; COGS = Beg + Purchases – Returns – End.
Perpetual	Records updated with every purchase and sale; annual count checks them.
FIFO	Oldest costs to COGS, newest stay. Rising prices: lowest COGS, highest income and ending inventory. Same result periodic or perpetual.
LIFO	Newest costs to COGS, oldest stay. Rising prices: highest COGS, lowest income. Periodic and perpetual can differ.

Parts Co.: 2,000 @5.00, buy 1,500 @5.40, sell 700, buy 400 @5.65, sell 1,300. FIFO: COGS 10,000, ending 10,360. LIFO: COGS 10,860, ending 9,500.

Property, plant and equipment

Cost includes	Purchase price, property taxes and import duties, transport in, installation and trial runs, interest during construction of a self-built asset (expensed once it is ready).
Depreciation	Systematic, rational allocation of cost over the periods that benefit. Land is not depreciated. See the formula sheet.
Accelerated vs straight-line	Early years: more expense, lower income, more accumulated depreciation, lower retained earnings. Gross fixed assets identical. Double-declining ignores salvage until the floor (final year 29,600 lands on 100,000).
Retirement	Sale, exchange, abandonment, involuntary conversion. Depreciate up to the date; remove cost and accumulated depreciation; proceeds – book value = gain or loss, shown below operating income within continuing operations.

Intangibles

Purchased	At cost including costs to prepare for use; if exchanged for other assets, at the more reliable fair value.
Internally created	Only direct costs such as legal fees. R&D is always expensed. Internally generated goodwill is never recorded; goodwill arises only when buying another company.
Finite life	Amortize cost minus residual over the useful life.
Indefinite life	No amortization; impairment test at least yearly.

Liabilities and financing

Leave out of current	A debt due within a year stays long-term if it will be settled from non-current assets (a dedicated fund) or the company has both the intent and the ability to refinance (actual refinancing before issuance, or a non-cancelable agreement).
Short-term sources	Trade credit · unsecured bank loans (promissory note; line of credit = available if the bank has funds; revolving credit = guaranteed) · commercial paper · secured loans on receivables or inventory (field warehousing, warehouse receipts), chattel mortgages, repurchase agreements · factoring · bankers' acceptances. A required compensating balance raises the effective interest rate.
Long-term sources	Term loans, leases, bonds (large reputable firms). Higher rates for the longer maturity.

B.1 Revenue recognition and equity

Five steps

Step	What to check	Worked number
1 Contract	Approved and committed · rights identifiable · payment terms identifiable · commercial substance · collection probable. Cash received before a valid contract or performance is a liability .	Company A receives 9,000 on May 1, delivers June 15: revenue June 15.
2 Performance obligations	One for each distinct good or service: the customer can benefit from it alone and it is separately identifiable.	Software + optional support = 2. Software + heavy customization = 1.
3 Transaction price	Variable amounts: expected value (many outcomes) or most likely amount (two outcomes). Financing component if payment is more than a year away: revenue at the cash price, rest is interest. Noncash consideration at fair value. Rebates and coupons reduce the price. Sales tax excluded.	$0.5 \times 360k + 0.3 \times 320k + 0.2 \times 300k = 336,000$. Sale for 13,310 in 3 years, cash price 10,000: revenue 10,000; interest 1,000, 1,100, 1,210. 5% probable rebate on 200,000: revenue 190,000.
4 Allocate	By relative standalone selling price; estimate it if not observable.	300,000 contract; SSP 250,000 + $3 \times 50,000 = 400,000 \rightarrow$ license 187,500, support 112,500.
5 Recognize	When control transfers (point in time) or as it transfers (over time). See the decision table on page 4. Progress over time by output (units, milestones) or input (cost-to-cost); straight-line if effort is even.	Support $112,500 \div 3 = 37,500$ per year.

Special cases

Right of return	Revenue only for units expected to stay sold; a refund liability (allowance for returns, contra-revenue); an asset for goods expected back. 40 units at 1,000, 4 expected back: net sales 36,000.
Repurchase agreement	Forward or call: repurchase price below original = lease ; at or above = financing . Put option: no incentive to exercise = sale with return; incentive and price below original = lease; at or above original and above market = financing.
Bill-and-hold	Revenue when control passes if: substantive reason, goods identified as the customer's, ready to ship, cannot be redirected.
Principal vs agent	Principal controls the good before transfer: revenue gross . Agent only arranges: revenue = commission . Agent signs: someone else fulfills, no inventory risk, no pricing power, paid by commission, no credit risk.
Consignment	Consignor keeps title and the inventory (plus freight); revenue only when the dealer reports a sale . Dealer records commission revenue and a payable. COGS on 10,000 of 30,000 cost with 1,500 freight = 10,500.

Expense recognition

Cause and effect (matching)	COGS, sales-return allowance, bad-debt allowance, warranty accrual
Systematic and rational allocation	Depreciation, prepaid insurance, interest and lease cost by passage of time
Immediate	Officers' salaries, general and administrative costs, a patent found worthless

Equity

Common stock	Votes, residual claim, pre-emptive right to keep ownership share. Par, no-par, book value, market value.
Preferred stock	Dividend preference and priority after debt; usually no vote; dividend not owed until declared; may be convertible, callable, sinking fund, participating. Cumulative : arrears (not a liability) must be paid before common. Noncumulative: current year only.
Treasury stock	Issued but not outstanding : no vote, no dividend. Contra-equity; reduces assets and equity. Bought to lift EPS and ROE, support the price, or supply shares for compensation.
Retained earnings	Free, or appropriated (restricted) by the board. Accumulated OCI holds gains and losses that bypass net income and retained earnings.
Prior-period adjustment	Correction of an earlier error (cash to accrual, missed expense): opening retained earnings, net of tax; not in this year's income.
Change in principle	Average cost to FIFO, completed-contract to percentage-of-completion: cumulative effect, net of tax, to opening retained earnings.
Change in estimate	Useful life, salvage, collectability, and depreciation method : prospective only.

B.2 Common-size analysis · B.3 Ratios

Vertical common-size

Each item as a % of the same year's base: **total assets** on the balance sheet, **net sales** on the income statement. Compares companies of different sizes, periods, or the industry average. "Advertising is 2% of sales."

Horizontal common-size (base year)

Each item as a % of its base-year value. Also called trend or variation analysis. Growth rate = index – 100. If sales go from 100 to 125, growth is 25%.

Largest change test: divide current by prior for every line. $500 \div 600 = 83\%$ is a 17% move and beats +10%, +8%, +5%.

Reading the Future Tech case

Signal	Meaning
Gross margin 25% → 22% → 19%	Cost of sales growing faster than prices
G&A 3% → 7% of sales	Overhead out of control
Inventory index 100 → 207 → 426	Stock piling up faster than sales
Cash 25% → 6% of assets; A/P 2% → 25%	Growth financed by stretching suppliers
Operating cash flow far below net income	Profit on paper, not in the bank

How to read each ratio family

Liquidity	Can it pay what is due soon? Quick ratio is stricter than current because inventory (a library, a department store) may take long to sell. Too high a current ratio means idle assets.
Leverage	How much is borrowed and can it service the interest? Creditors like low debt ratios; cyclical firms keep them low; owners accept more because debt magnifies returns. Interest is tax deductible, dividends are not. Operating leverage = fixed costs magnify the effect of sales swings on operating income.
Activity	How fast do assets turn into sales and cash? Shorter operating and cash cycles are better. Too low a DSO can mean credit terms so tight that sales are lost.
Profitability	Margins per dollar of sales; ROA is earning power of the assets; ROE is return to owners. ROE above ROA means borrowed money earned more than it cost. If operating margin falls while gross margin holds, the problem is other operating costs.

Fictitious inventory inflates total assets: **total asset turnover falls** unexpectedly. The current ratio would go up, not down. DSO and P/E are unaffected.

Average balances: $(\text{beginning} + \text{ending}) \div 2$. Beechwood's equity includes the reserve for bond retirement but not the bonds: $(467,000 + 534,000) \div 2 = 500,500$; $\text{ROE} = 96,000 \div 500,500 = 19.2\%$.

From a quick ratio to inventory: $\text{current assets} = \text{NWC} + \text{current liabilities}$; $\text{inventory} = \text{current assets} - \text{quick ratio} \times \text{current liabilities}$.